

3728

Test - 2

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V. Good!
Sk mihira
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1) a) $(101101)_2$

$$\therefore 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 32 + 0 + 8 + 4 + 0 + 1$$

$$= (45)_{10}$$

$$\therefore \text{Ans} \rightarrow (45)_{10}$$

b) $(1110101)_2$

$$= 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 64 + 32 + 16 + 0 + 4 + 0 + 1$$

$$= 96 + 7$$

$$= (117)_{10}$$

$$\therefore \text{Ans} \rightarrow (117)_{10}$$

c) $(110101.101)_2$

$$= \begin{matrix} 1 \times 2^5 & + & 1 \times 2^4 & + & 0 \times 2^3 & + & 1 \times 2^2 & + & 0 \times 2^1 & + & 1 \times 2^0 \\ 1 \times 2^{-1} & + & 0 \times 2^{-2} & + & 1 \times 2^{-3} \end{matrix}$$

$$= 32 + 16 + 0 + 4 + 0 + 1 + \frac{1}{2} + 0 + \frac{1}{8}$$

$$= 53 + \frac{41}{8}$$

$$= 53 + \frac{5}{8}$$

$$= \frac{424 + 5}{8}$$

$$= \frac{429}{8}$$

$$= (53.625)_{10}$$

$$\therefore \text{Ans} \Rightarrow (53.625)_{10} \checkmark$$

$$\text{Prob 1) } (10011.011)_2$$

$$= 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 16 + 0 + 0 + 2 + 1 + 0 + \frac{1}{2} + \frac{1}{8}$$

$$= 19 + \frac{2+1}{8}$$

$$= 19 + \frac{3}{8}$$

$$= \frac{152 + 3}{8}$$

$$= \frac{155}{8}$$

$$= (19.375)_{10}$$

$$\therefore \text{Ans} \Rightarrow (19.375)_{10} \checkmark$$

R.W

$$8 \overline{) 53.625}$$

$$\begin{array}{r} 429 \\ -40 \downarrow \\ \hline 29 \end{array}$$

$$\begin{array}{r} 24 \\ -24 \\ \hline 50 \end{array}$$

$$\begin{array}{r} 48 \\ -48 \\ \hline 20 \end{array}$$

$$20$$

$$\begin{array}{r} 16 \\ -16 \\ \hline 40 \end{array}$$

$$\begin{array}{r} 40 \\ -40 \\ \hline 0 \end{array}$$

$$8 \overline{) 19.375}$$

$$\begin{array}{r} 155 \\ -84 \downarrow \\ \hline 75 \end{array}$$

$$\begin{array}{r} 72 \\ -72 \\ \hline 30 \end{array}$$

$$\begin{array}{r} 24 \\ -24 \\ \hline 60 \end{array}$$

$$\begin{array}{r} 56 \\ -56 \\ \hline 40 \end{array}$$

$$\begin{array}{r} 40 \\ -40 \\ \hline 0 \end{array}$$

(4)

$$2) a) \quad (57)_{10}$$

$$\begin{array}{r|l} 2 & 57 \\ \hline 2 & 28 \quad 1 \\ 2 & 14 \quad 0 \\ 2 & 7 \quad 0 \\ 2 & 3 \quad 1 \\ & 1 \quad 1 \end{array}$$

$$\therefore \text{Ans} \rightarrow (111001)_2$$

$$b) \quad (125)_{10}$$

$$\begin{array}{r|l} 2 & 125 \\ \hline 2 & 62 \quad 1 \\ 2 & 31 \quad 0 \\ 2 & 15 \quad 1 \\ 2 & 7 \quad 1 \\ 2 & 3 \quad 1 \\ & 1 \quad 1 \end{array}$$

$$\therefore \text{Ans} \rightarrow (1111101)_2$$

$$c) \quad (45.625)_{10}$$

$$\begin{array}{r|l} 2 & 45 \\ \hline 2 & 22 \quad 1 \\ 2 & 11 \quad 0 \\ 2 & 5 \quad 1 \\ 2 & 2 \quad 1 \\ & 1 \quad 0 \end{array}$$

$$\begin{array}{r} 625 \\ \times 2 \\ \hline (1)2 \quad 50 \\ \times 2 \\ \hline (0)500 \\ \times 2 \\ \hline (1)000 \end{array}$$

$$\therefore \text{Ans} \rightarrow (101101.101)_2$$

$$d) (23.375)_{10}$$

$$\begin{array}{r} 2 \overline{) 23} \\ 2 \overline{) 11} \quad 1 \\ 2 \overline{) 5} \quad 1 \\ 2 \overline{) 2} \quad 1 \\ 1 \quad 0 \end{array}$$

$$\begin{array}{r} 1'375 \\ \times 2 \\ \hline (0)750 \\ \times 2 \\ \hline (1)500 \\ \times 2 \\ \hline (1)000 \end{array}$$

$$\therefore \text{Ans} \rightarrow (10111.011)_2$$

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$$3)a) (\overline{110111101})_2$$

$$= (675)_8$$

$$b) (572)_8$$

$$= (101111010)_2$$

$$c) (\overline{101101011})_2$$

$$= (16B)_{16}$$

$$d) (9AF)_{16}$$

$$= (100110101111)_2$$

~~$$e) (156)_{10} = (\overline{11011110})_2 = (BE)_{16}$$~~

e) $(156)_{10}$

$$\begin{array}{r|l}
 2 & 156 \\
 \hline
 2 & 78 \quad 0 \\
 2 & 39 \quad 0 \\
 2 & 19 \quad 1 \\
 2 & 9 \quad 1 \\
 2 & 4 \quad 1 \\
 2 & 2 \quad 0 \\
 & 1 \quad 0
 \end{array}$$

$$\begin{array}{r|l}
 16 & 156 \\
 \hline
 16 & 9 \dots C \uparrow \\
 & 0 \quad 9 \uparrow
 \end{array}$$

$$(156)_{10} = (9C)_{16}$$

R-W

$$\begin{array}{r}
 2^3 \\
 512 \\
 256 \\
 128 \\
 64 \\
 32 \\
 8 \\
 + 1 \\
 \hline
 1001
 \end{array}$$

$$\therefore (156)_{10} = (10011100)_2 \xrightarrow{12} C$$

$$= (9B)_{16}$$

Ans $\Rightarrow (9B)_{16}$ X

f) $(3E9)_{16}$

$$\begin{aligned}
 & 3 \times 16^2 + 14 \times 16^1 + 9 \times 16^0 \\
 & = 3 \times 256 + 224 + 9 = 768 + 233 \\
 & = 1001
 \end{aligned}$$

$$= \text{decimal } (1111101001)_2$$

$$= 1 \times 2^9 + 1 \times 2^8 + 1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 512 + 256 + 128 + 64 + 32 + 0 + 8 + 0 + 0 + 1$$

$$= (1001)_{10}$$

Ans $\Rightarrow (1001)_{10}$ ✓

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$$4) a) (4D7)_{16}$$

$$= (\overline{1001101011})_2$$

$$= (2327)_8$$

$$\therefore \text{Ans} \rightarrow \cancel{2327} (2327)_8$$

$$b) (745)_8$$

$$= \cancel{111010101}_2$$

$$= \boxed{\cancel{111010101}} (1D5)_{16}$$

$$\therefore \text{Ans} \rightarrow \boxed{\cancel{111010101}} (1D5)_{16}$$

$$c) (A3F)_{16}$$

$$= (\overline{101000011111})_2$$

$$= (5077)_8$$

$$\therefore \text{Ans} \rightarrow (5077)_8$$

~~7) a)~~

7) a)

$$(1011)_2 + (110)_2$$

$$\begin{array}{r} 1011 \\ \times 110 \\ \hline 10000 \end{array}$$

$$1011X$$

$$\begin{array}{r} + 1011XX \\ \hline 1000010 \end{array}$$

$$\therefore \text{Ans} \rightarrow (1000010)_2$$

$$b) (111100)_2 \div (101)_2$$

$$\therefore (111100)_2 = 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0$$

$$= 32 + 16 + 8 + 4 + 0 + 0$$

$$= (60)_{10}$$

$$\therefore (101)_2 = 1 \times 2^2 + 0 \times 2 + 1 \times 2^0$$

$$= 4 + 0 + 1$$

$$= (5)_{10}$$

$$\therefore (111100)_2 \div (101)_2$$

$$= (60)_{10} \div (5)_{10}$$

$$\begin{array}{r} 12 \\ 5 \overline{) 60} \\ \underline{-50} \\ 10 \\ \underline{-10} \\ 0 \end{array}$$

$$\begin{array}{r} 101 \overline{) 111100} \\ \underline{101} \\ 0101 \\ \underline{101} \\ 0000 \\ \underline{0} \\ 00000 \\ \underline{00000} \\ 00000 \\ \underline{00000} \\ 00000 \\ \underline{00000} \\ 00000 \\ \underline{00000} \\ 00000 \end{array}$$

6

~~Ans~~

$$\therefore (60)_{10} \div (5)_{10} = (12)_{10}$$

$$\therefore (12)_{10}$$

$$\text{Ans: } 1100$$

$$\begin{array}{r} 2 \overline{) 12} \\ 2 \overline{) 6} \\ 2 \overline{) 3} \\ 1 \end{array}$$

$$\therefore \text{Ans} \Rightarrow (1100)_2$$

[don't convert into decimal and divide]
 do it directly

$$\therefore \text{Ans} \Rightarrow (1010)_2$$

679) $(101101)_2 - (011011)_2$

$\therefore$ 1's Complement of second no. = $(100100)_2$

$\therefore$ Adding 1st and 1's Complement of 2nd no.:

$$\begin{array}{r} 101101 \\ + 100100 \\ \hline (1)101001 \end{array}$$

$\therefore$ Adding ~~Do~~ Extra Digit:

$$\begin{array}{r} 010001 \\ + 1 \\ \hline 010010 \end{array}$$

$\therefore$ Ans $\rightarrow (10010)_2$ ✓

b) $(1111000)_2 - (0110101)_2$

$\therefore$ 1's Complement of 2nd no. = $(1001010)_2$

$\therefore$ 2's Complement of 2nd no. = 1001010

$$\begin{array}{r} 1111000 \\ + 1001010 \\ \hline (1)1000110 \end{array}$$

$\therefore$ Adding 1st no. and 2's Complement of 2nd no.:

$$\begin{array}{r} 1111000 \\ + 1001011 \\ \hline (1)1000111 \end{array}$$

$\therefore$ Discarding Extra Digit = $(1000011)_2$

$\therefore$ Ans $\rightarrow (1000011)_2$ ✓

4

$$5) a) (101101)_2 + (110111)_2$$

$$\begin{array}{r} 101101 \\ + 110111 \\ \hline 1100100 \end{array}$$

$$\therefore \text{Ans} \rightarrow (1100100)_2 \quad \checkmark$$

$$b) (1110101)_2 + (1011110)_2$$

$$\begin{array}{r} 1110101 \\ + 1011110 \\ \hline 11010011 \end{array}$$

4

$$\therefore \text{Ans} \rightarrow (11010011)_2 \quad \checkmark$$

$$8) a) (A7)_{16} + (3D)_{16}$$

$$\begin{array}{r} A7 \\ + 3D \\ \hline E4 \end{array}$$

$$\therefore \text{Ans} \rightarrow (E4)_{16}$$

$$b) 3(4)_{16} - (5B)_{16}$$

$$\begin{array}{r} 3^B \quad 4^{20} \\ - 5B \\ \hline 69 \end{array}$$

$$\therefore \text{Ans} \rightarrow (69)_{16} \quad \checkmark$$

$$c) (2A)_{16} + (4)_{16}$$

$$\begin{array}{r} 2A \\ + 4 \\ \hline A8 \end{array}$$

$$\therefore \text{Ans} \Rightarrow (A8)_{16}$$

$$d) (9C)_{16} \div (3)_{16}$$

$$\begin{aligned} \therefore (9C)_{16} &= (9 \times 16 + C \times 16^0) \\ &= 144 + 12 \\ &= (156)_{10} \end{aligned}$$

$$\begin{aligned} (3)_{16} &= 3 \times 16^0 \\ &= (3)_{10} \end{aligned}$$

$$\begin{aligned} \therefore (9C)_{16} \div (3)_{16} \\ = (156)_{10} \div (3)_{10} \end{aligned}$$

$$\begin{array}{r} 3 \overline{) 9C} 34 \\ \underline{9} \\ 0C \\ \underline{0C} \\ 0 \end{array}$$

$$\begin{array}{r} 52 \overline{) 156} \\ \underline{156} \\ 0 \end{array}$$

$$\text{Ans: } (34)_{16}$$

$$\therefore (156)_{10} \div (3)_{10} = (52)_{10}$$

$$\therefore (52)_{10}$$

$$\begin{array}{r} 2 \overline{) 52} \\ 2 \overline{) 26} \quad 0 \\ 2 \overline{) 13} \quad 6 \\ 2 \overline{) 6} \quad 1 \\ 2 \overline{) 3} \quad 0 \\ 1 \quad 0 \quad 1 \end{array}$$

Do direct division!

No need to do the conversions.

(8)

$$\therefore (52)_{10} = (110100)_2 = (34)_{16}$$

~~Ans $\Rightarrow 1101$~~

$$\therefore \text{Ans} \Rightarrow (34)_{16} \quad \checkmark$$

$$9) a) (657)_8 + (245)_8$$

$$\begin{array}{r} ^1 6 ^1 5 7 \\ + ^1 2 ^1 4 5 \\ \hline ^1 1 ^1 2 4 \end{array}$$

$$\therefore \text{Ans} \Rightarrow (1124)_8 \quad \checkmark$$

$$b) (732)_8 - (165)_8$$

$$\begin{array}{r} ^6 7 ^{10} 3 ^{10} 2 \\ - ^1 1 ^1 6 ^1 5 \\ \hline ^1 5 ^1 4 ^1 5 \end{array}$$

$$\therefore \text{Ans} \Rightarrow (545)_8 \quad \checkmark$$

R.W

$\begin{array}{r} 64 \\ \times 6 \\ \hline 384 \\ 40 \\ \hline 431 \end{array}$	$\begin{array}{r} 128 \\ 32 \\ 5 \\ \hline 165 \end{array}$
$\begin{array}{r} 364 \\ \times 8 \\ \hline 572 \end{array}$	$\begin{array}{r} 431 \\ 595 \\ \hline 596 \end{array}$
$\begin{array}{r} 64 \\ \times 7 \\ \hline 448 \\ 24 \\ \hline 472 \end{array}$	$\begin{array}{r} 512 \\ 48 \\ 5 \\ \hline 565 \end{array}$
$\begin{array}{r} 64 \\ \times 5 \\ \hline 320 \\ 48 \\ 5 \\ \hline 325 \end{array}$	$\begin{array}{r} 164 \\ 48 \\ 5 \\ \hline 117 \end{array}$
$\begin{array}{r} 64 \\ \times 3 \\ \hline 192 \\ 24 \\ \hline 216 \end{array}$	$\begin{array}{r} 596 \\ 5 \\ \hline 596 \end{array}$
$\begin{array}{r} 64 \\ \times 2 \\ \hline 128 \\ 48 \\ 5 \\ \hline 133 \end{array}$	$\begin{array}{r} 596 \\ 5 \\ \hline 596 \end{array}$
$\begin{array}{r} 64 \\ \times 1 \\ \hline 64 \\ 48 \\ 5 \\ \hline 117 \end{array}$	$\begin{array}{r} 596 \\ 5 \\ \hline 596 \end{array}$

$$c) (37)_8 + (5)_8$$

$$\begin{array}{r} 37 \\ + 5 \\ \hline 233 \end{array}$$

$$\therefore \text{Ans} \Rightarrow (233)_8 \quad \checkmark$$

$$d) (754)_8 \div (6)_8$$

$$\begin{aligned} \therefore (754)_8 &= (7 \times 64) + (5 \times 8) + (4 \times 1) \\ &= (492)_{10} \end{aligned}$$

$$\therefore (6)_8 = (6 + 8^0) = (6)_{10}$$

$$\therefore (754)_8 \div (6)_8$$

$$= (492)_{10} \div (6)_{10}$$

$$\begin{array}{r} 82 \\ 6 \overline{) 492} \\ \underline{-48} \\ 12 \\ \underline{-12} \\ 0 \end{array}$$

$$\begin{array}{r} 122 \\ 6 \overline{) 754} \\ \underline{6} \\ 15 \\ \underline{14} \\ 14 \\ \underline{14} \\ 0 \end{array}$$

$$\therefore (492)_{10} \div (6)_{10} = (82)_{10}$$

$$\text{Ans: } (122)_8$$

R.W

$$264$$

$$\begin{array}{r} 7 \\ 448 \end{array}$$

$$40$$

$$4$$

$$492$$

$$24$$

$$7$$

$$31$$

$$85$$

$$155$$

$$128$$

$$24$$

$$3$$

$$5$$

$$492$$

$$40$$

$$9$$

$$492$$

$$64$$

$$16$$

$$2$$

$$82$$

$$\begin{array}{r} 1 \\ 8 \overline{) 12} \\ \underline{8} \\ 4 \end{array}$$

$$(82)_{10}$$

$$\begin{array}{r}
 2 \overline{) 82} \\
 \underline{2 \overline{) 41}} \quad 0 \\
 \underline{2 \overline{) 20}} \quad 1 \\
 \underline{2 \overline{) 10}} \quad 0 \\
 \underline{2 \overline{) 3}} \quad 0 \\
 \underline{2 \overline{) 2}} \quad 1 \\
 \quad \quad 1 \quad 0
 \end{array}$$

$$(82)_{10} = (1010010)_2 = (122)_8$$

$$\text{Ans} \Rightarrow (122)_8$$

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